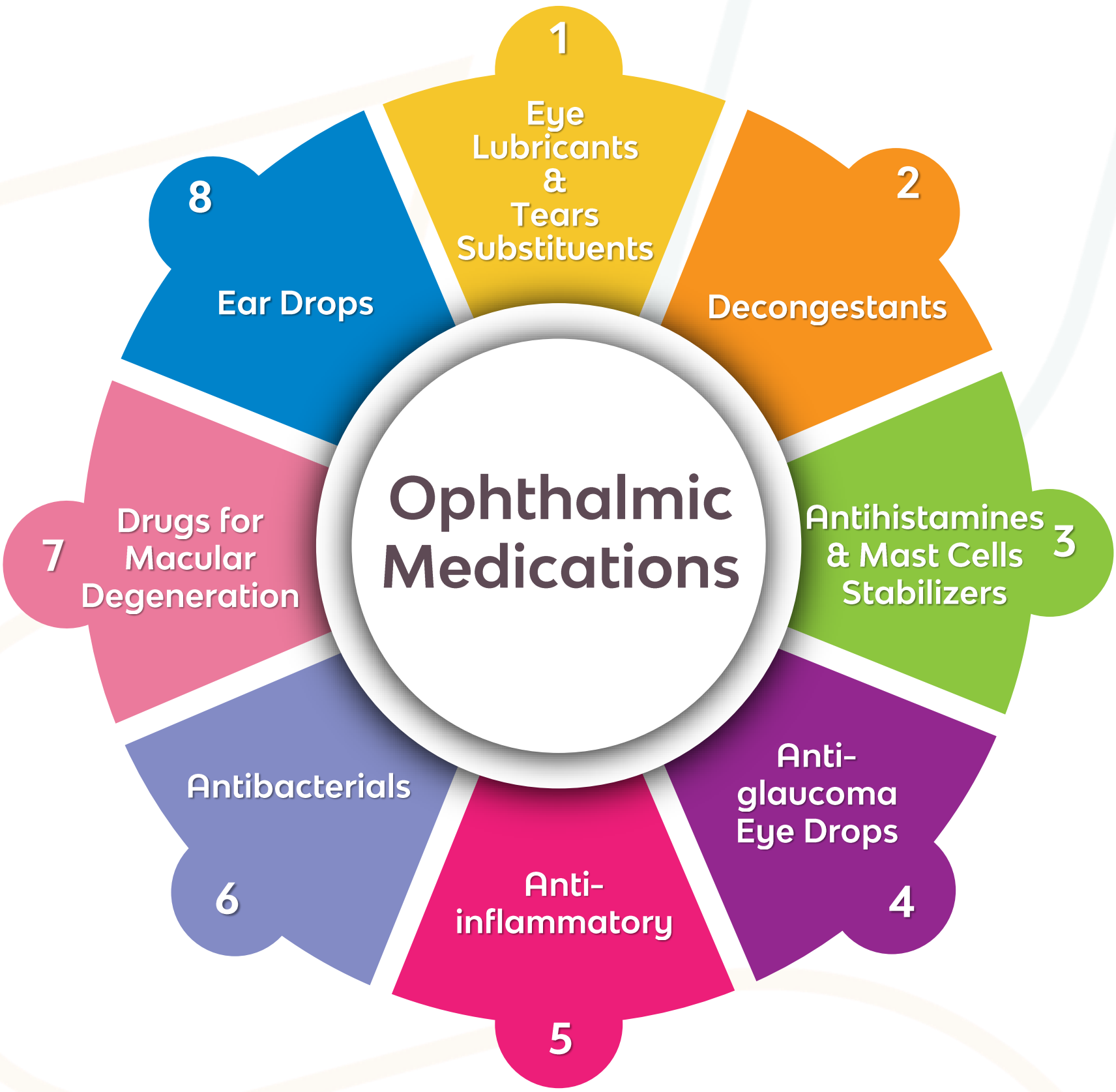
Eye & Ear Part 1



Talent Management Department

صيدليـة المتحـدة
United Pharmacy

Eye & Ear drugs classification



To day we will learn about

01 Eye Lubricants & Tears substituents

02 Decongestants

03 Antihistamines & Mast Cells Stabilizers

04 Anti-glaucoma Eye Drops

Eye Lubricants & Tears substituents

These are used to relieve dry eyes by moisturizing the surface of the eye, mimicking natural tears, and protecting the cornea. They are commonly used in: Dry eye syndrome, Environmental irritation (dust, wind, screen use), Post-surgery or contact lens-related dryness

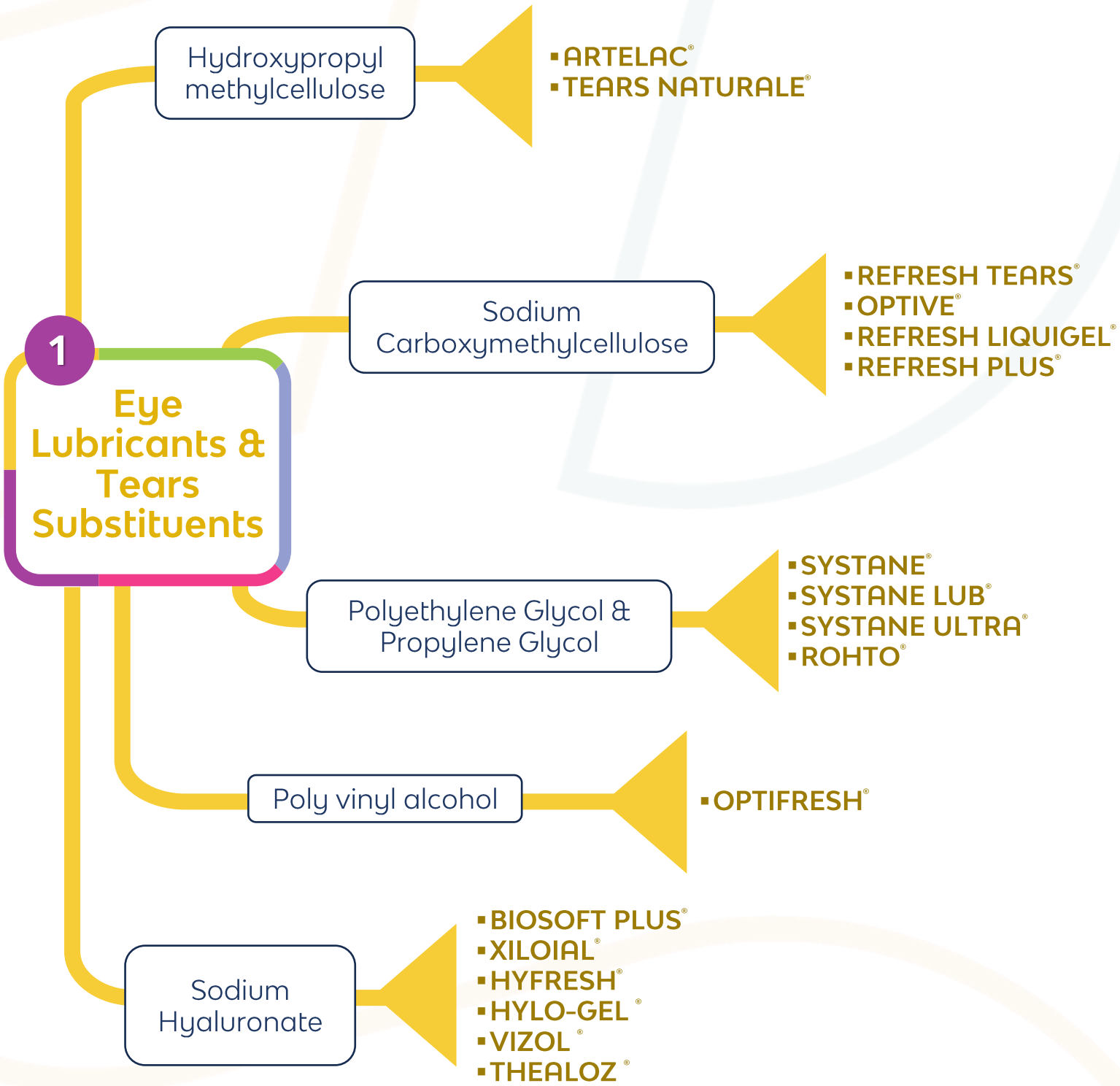
1. Hydroxypropyl methylcellulose (HPMC) ✅
- Forms a protective film; increases tear retention time.

2. Sodium carboxymethylcellulose (CMC) ✅
- High viscosity → long-lasting effect; commonly used in moderate to severe dry eye.

3. Polyethylene glycol + Propylene glycol ✅
- Dual action to lubricate and protect; commonly found in preservative-free formulas.

4. Polyvinyl alcohol (PVA) ✅
- Low-viscosity tear substitute; less sticky, better for mild symptoms.

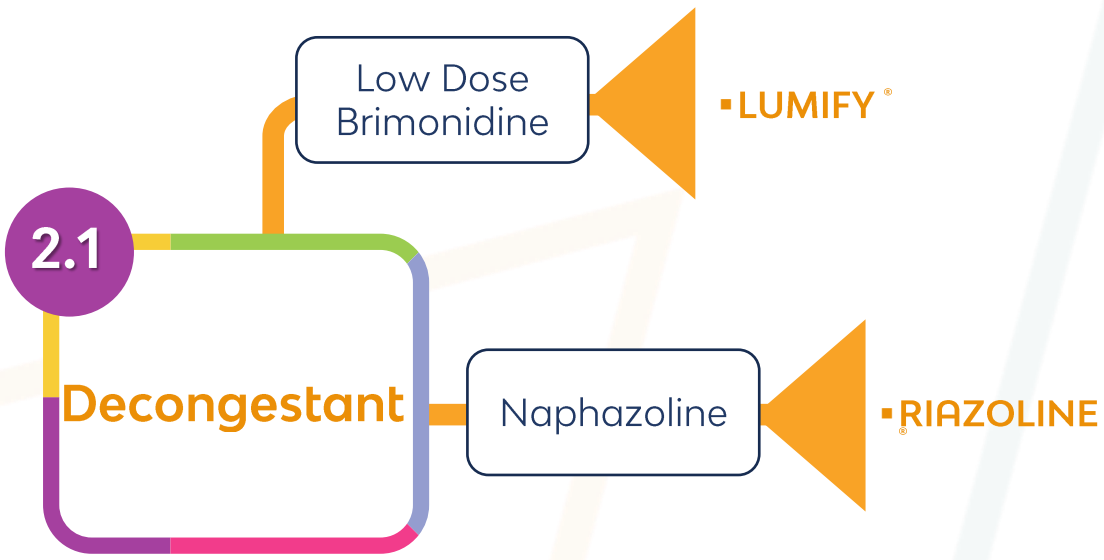
5. Sodium hyaluronate ✅
- Highly moisturizing; binds water strongly; excellent for severe dry eyes, with healing properties.

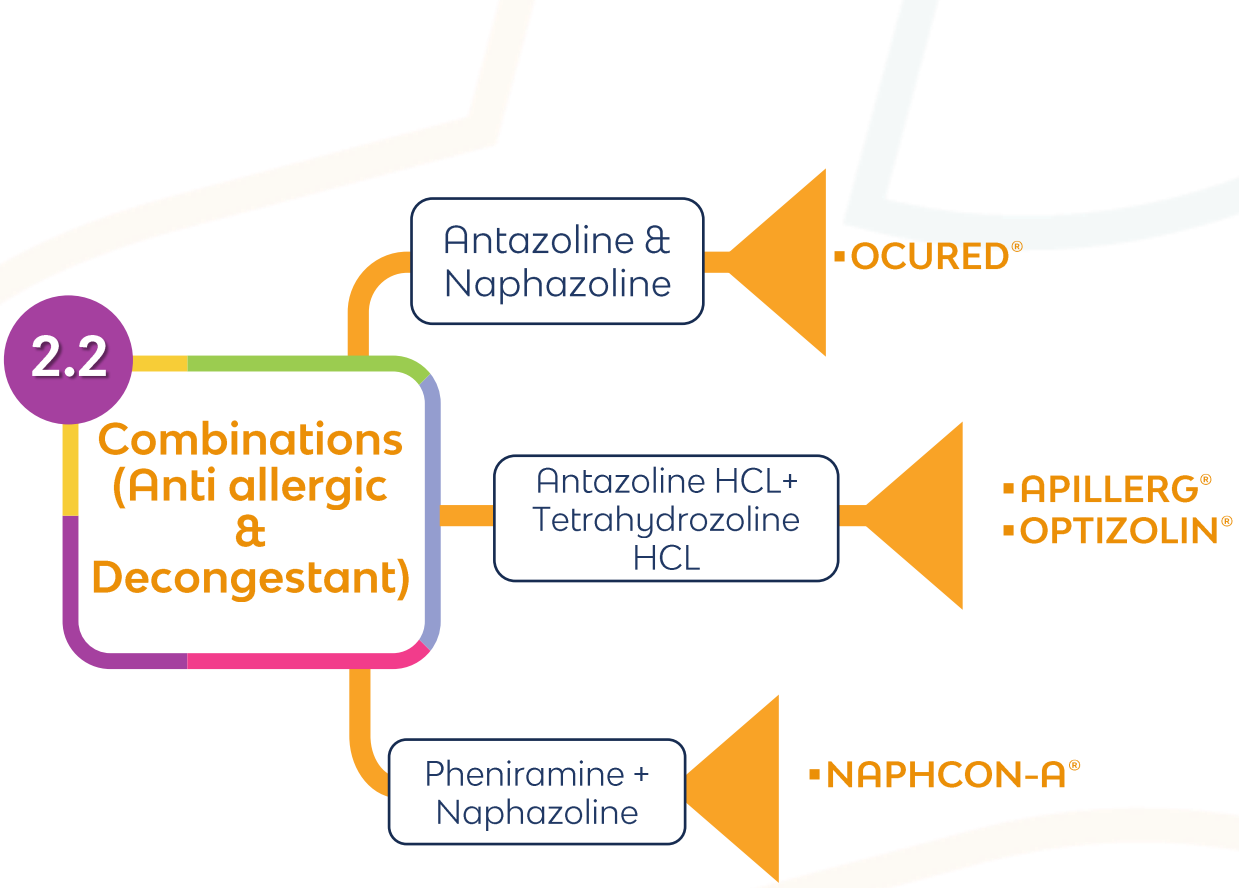

1
Eye Lubricants & Tears Substituents
Hydroxypropyl methylcellulose
ARTELAC®
TEARS NATURALE®
Sodium Carboxymethylcellulose
REFRESH TEARS®
OPTIVE®
REFRESH LIQUIGEL®
REFRESH PLUS®
Polyethylene Glycol & Propylene Glycol
SYSTANE®
SYSTANE LUB®
SYSTANE ULTRA®
ROHTO®
Poly vinyl alcohol
OPTIFRESH®
Sodium Hyaluronate
BIOSOFT PLUS®
XILOIAL®
HYFRESH®
HYLO-GEL®
VIZOL®
THEALOZ®

Decongestants

These eye drops are used to relieve eye redness caused by minor irritation, allergies, or environmental factors (like dust).
They work by constricting blood vessels on the eye surface.

- Act as vasoconstrictors → Narrow the conjunctival blood vessels → reduce redness.


2.1
Decongestant
Low Dose Brimonidine
LUMIFY®
Naphazoline
RIAZOLINE®


2.2
Combinations (Anti allergic & Decongestant)
Antazoline & Naphazoline
OCURED®
Antazoline HCL+ Tetrahydrozoline HCL
APILLERG®
OPTIZOLIN®
Pheniramine + Naphazoline
NAPHCON-A®

Antihistamines & Mast Cells Stabilizers

Used to treat allergic conjunctivitis (itchy, red, watery eyes caused by allergens like dust, pollen, or pets). They work by blocking histamine and/or preventing histamine release from mast cells.

1. Antihistamines ✅

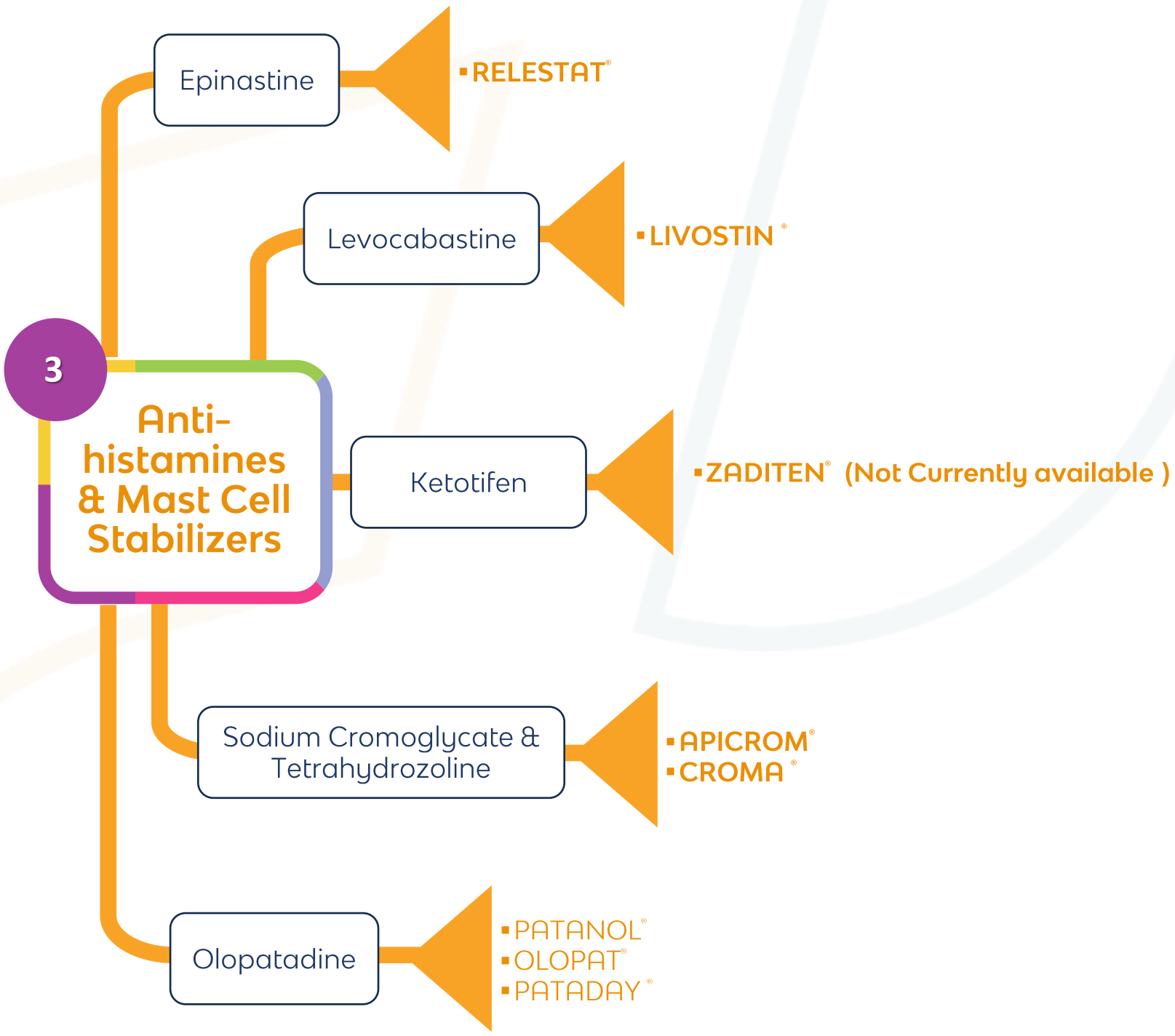
- They Block H1 histamine receptors → reduce itching and redness, e.g. Levocabastine.

2. Mast Cell Stabilizers ✅

- Prevent mast cells from releasing histamine and other allergy mediators, e.g. Sodium Cromoglycate.

3. Dual action drugs ✅

- They have both H1 blocker activity and mast cells stabilizing activities, e.g. Epinastine, Ketotifen & Olopatadine.


3
Anti-histamines & Mast Cell Stabilizers
Epinastine
RELESTAT®
Levocabastine
LIVOSTIN ®
Ketotifen
ZADITEN® (Not Currently available)
Sodium Cromoglycate & Tetrahydrozoline
APICROM®
CROMA ®
Olopatadine
PATANOL®
OLOPAT®
PATADAY ®

Anti-Glaucoma Eye Drops

Used to lower intraocular pressure (IOP) in glaucoma and ocular hypertension to prevent optic nerve damage and vision loss, Reduce IOP by either: Decreasing aqueous humor production or Increasing aqueous humor outflow.

1. Decreasing aqueous humor production ✅

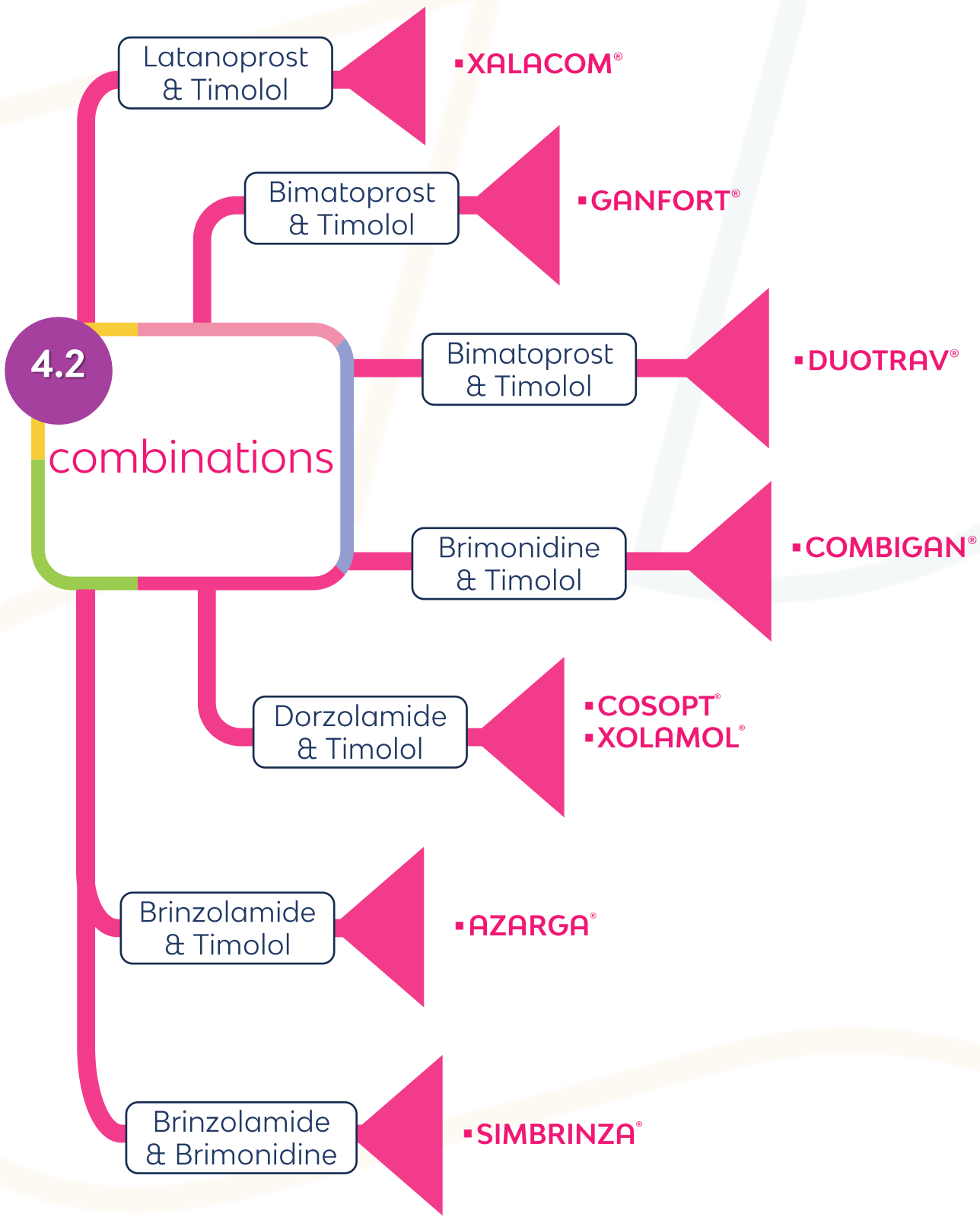
- α2 agonists.
- β-Blockers.
- Carbonic Anhydrase Inhibitors.

2. Increasing aqueous humor outflow ✅

- Cholinomimetics.
- Prostaglandin analogues.

4 Anti-Glaucoma Eye Drops

- Cholinomimetics
  - Pilocarpine
    - APICARPIN®
- Prostaglandin F2α analogues
  - Latanoprost
    - XALATAN®
    - LATANO®
  - Bimatoprost
    - LUMIGAN®
  - Travoprost
    - TRAVATAN®
- α2 agonists
  - Brimonidine
    - ALPHAGAN-P®
    - BRIMO-P®
- β-Blockers
  - Timolol
    - APIMOL®
  - Betaxolol
    - BETOPTIC®
- Carbonic anhydrase inhibitors
  - Brinzolamide
    - AZOPT®
  - Dorzolamide
    - TRUSOPT®
    - XOLA®


4.2
combinations
Latanoprost & Timolol
XALACOM®
Bimatoprost & Timolol
GANFORT®
Bimatoprost & Timolol
DUOTRAV®
Brimonidine & Timolol
COMBIGAN®
Dorzolamide & Timolol
COSOPT®
XOLAMOL®
Brinzolamide & Timolol
AZARGA®
Brinzolamide & Brimonidine
SIMBRINZA®

الآن حان دور الأنشطة
اضغط على الزر للبدء



ابدأ النشاط